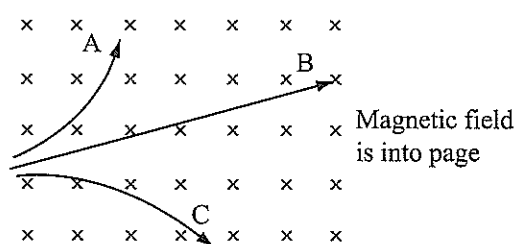


Problem Solving and Calculations

Set 14: Charged Particles in Magnetic Fields

Notes

- Explain what we mean by the term *magnetic field*.
 - Sketch the shape of the magnetic field surrounding the Earth. Indicate clearly the geographic and magnetic poles, and show the direction of the field.
 - A short straight length of steel wire that has been stroked from end to end with a magnet, and a short straight length of copper wire carrying a current, each produce a magnetic field. Describe how these fields *differ*.
 - How could the copper wire be made to produce a field more like the field produced by the steel wire?
- Sketch the magnetic field distribution for each of the following arrangements, indicating any null points:
 - bar magnet,
 - two bar magnets in line, with opposite poles facing each other and separated by 50 mm,
 - two bar magnets in line, with similar poles facing each other and separated by 50 mm,
 - two bar magnets placed parallel to each other, about 50 mm apart, and with opposite poles adjacent,
 - two bar magnets in line, separated by 80 mm, with opposite poles facing and with a small iron washer placed midway between them.
- Draw the magnetic field pattern for each of the following:
 - a single straight current-carrying copper wire, and
 - a current-carrying circular coil of wire.
- An electron, travelling at a constant speed, enters a region of uniform magnetic field. Describe the subsequent motion of the electron if the field direction is:
 - parallel to the electron's direction of motion;
 - perpendicular to the electron's direction of motion.
- Particles A, B and C follow the paths shown in the diagram as they pass through a magnetic field. What conclusions can you draw about the charges on particles A, B and C?

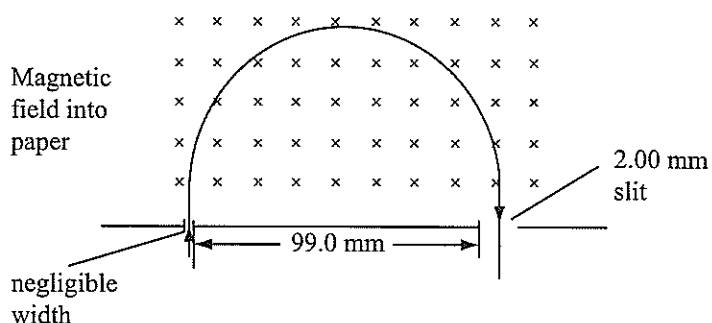


- A proton of mass m is emitted from the Sun and enters the Earth's magnetic field with a velocity v . Assume that this magnetic field has a magnitude of B , is uniform and at right angles to the direction of travel of the proton.
 - Using the variables listed above, write an expression for the force F experienced by the proton.
 - Hence deduce the subsequent path of the proton. Use a labelled diagram and explain clearly why the path has its particular shape.
 - This path is a repeated one. Derive an expression for the rate at which it is repeated (that is, derive an expression for the frequency).
 - If you could measure this frequency, what information (if any) would it give you about the speed of the proton and the strength of the Earth's magnetic field?

Problem Solving and Calculations 14

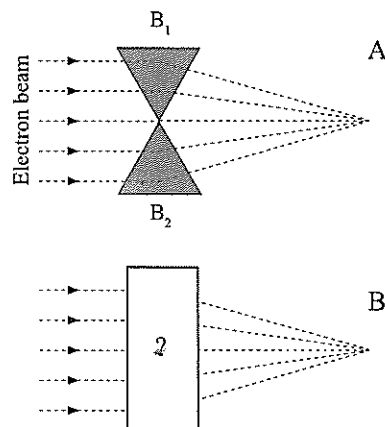
Set 14: Charged Particles in Magnetic Fields

7. A beam of particles, each with a charge of $+1.60 \times 10^{-19} \text{ C}$ and a velocity of $5.00 \times 10^6 \text{ m s}^{-1}$, is directed through a narrow aperture into a uniform magnetic field of strength 10.0 T as shown at right. The particles move in a semi-circular path and leave the magnetic field through a slit of width 2.00 mm as shown.

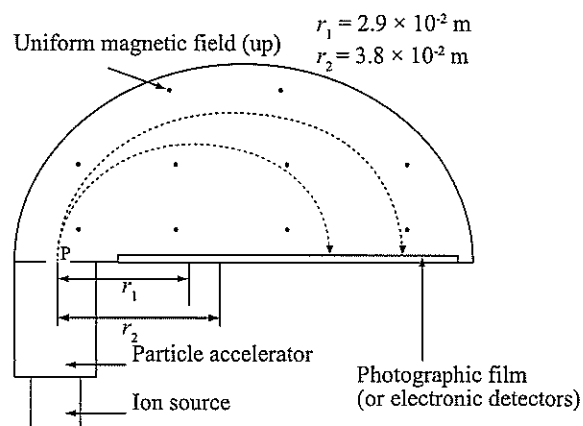


Find the range of masses of particles in the beam that could pass through this slit.

8. A proton moving at $1.00 \times 10^4 \text{ m s}^{-1}$ in a uniform magnetic field of $2.50 \mu\text{T}$ travels at right angles to the field, and remains in the field.
- Determine the radius of the proton's path in this field.
 - In this field, how long does the proton take to complete one revolution?
 - Explain how the time for one revolution might change if the magnetic field strength is increased.
 - Explain how the time for one revolution might change if the proton's initial velocity is greater.
9. An electron microscope uses a "magnetic lens" to focus a wide beam of electrons to a point, as shown in diagram A. In this case, all the electrons have the same speed.
- What would be the directions of magnetic fields B_1 and B_2 ?
 - The electron velocity is $1.5 \times 10^6 \text{ m s}^{-1}$ and the magnetic lens has field strength 0.10 T . Calculate the deflecting force.
 - The triangular shape of the magnetic field deviates the outer electrons more than the inner electrons. Explain.
 - If the magnetic field was not triangular but had the shape shown in diagram B, how could you achieve different amounts of deviation?
10. A mass spectrometer allows researchers to determine the charge to mass ratio $\frac{q}{m}$ of the various components of a mixture of isotopes or fragments of a molecule.



- Derive an expression relating $\frac{q}{m}$ to the radius, velocity and magnetic field strength.
- A mixture of helium isotopes produces lines as shown at right. If their velocity on entering at P is $2.20 \times 10^5 \text{ m s}^{-1}$ and the field strength is 0.120 T , find the charge to mass ratio of these isotopes.
- Two helium isotopes are ${}^4_2\text{He}^{2+}$ and ${}^3_2\text{He}^{2+}$. Could these isotopes produce the lines shown? Explain.
- In a similar experiment a sample of oxygen forms $+1$ ions that enter the mass spectrometer at $4.5 \times 10^4 \text{ m s}^{-1}$. The magnetic field strength is unchanged. The technician identifies three particles at radii 62.0 mm , 66.4 mm , and 70.1 mm . Identify the three isotopes of oxygen in her sample.



Problem Solving and Calculations

Set 15: Charged Particles in Electric and Magnetic Fields

Notes

- Moving charges may be deflected by electric or by magnetic fields. Describe the subsequent motion of a proton that enters
 - a uniform electric field parallel to the field lines
 - a uniform magnetic field parallel to the field lines
 - a uniform electric field at right angles to the field lines
 - a uniform magnetic field at right angles to the field lines
- An electron is fired vertically downwards between two vertical, parallel, charged metal plates. The West plate has a positive potential with respect to the East plate. To exactly balance the effect of the electric field, what must be the direction of a magnetic field in this region?
- A region of space contains a magnetic field of intensity 24.5 millitesla. A proton travelling at $4.50 \times 10^6 \text{ m s}^{-1}$ enters this field at right angles.
 - Calculate the strength of an electric field in the same region of space that would allow the proton to pass through undeflected.
 - Would an electron, travelling at the same speed and also entering the region at right angles to the two fields, pass through undeflected? Explain.
- A narrow electron beam enters a region at a speed of $6.00 \times 10^6 \text{ m s}^{-1}$ and is suddenly influenced by a magnetic field.
 - Describe and explain the effect on the electrons in the beam if the field is perpendicular to the beam.
 - Describe and explain the effect on the electrons in the beam if the field is parallel to the beam.
 - The electron beam leaves the magnetic field and enters a region in which there exists an electric field of strength $1.00 \times 10^3 \text{ V m}^{-1}$. Calculate the path of the electrons in the beam if the field is in the same direction as the beam.
- An electron, initially at rest, is accelerated through a potential difference of 15.0 kV. It is then allowed to circulate at right angles to a uniform magnetic field of strength 2.35 T.
 - Calculate the electron's final speed before entering the magnetic field.
 - Determine the electron's path radius in the magnetic field.
 - Assuming a completely circular path, calculate the electron's time of rotation.
- A proton moves in a circular path of radius 50.0 mm at right angles to a uniform magnetic field of strength 1.50 T.
 - Calculate the velocity required for the proton to maintain a circular orbit while in the field.
 - Determine the frequency and period of the proton's rotation.
 - Calculate the potential difference through which the proton was accelerated in order to have this speed.

Problem Solving and Calculations

Set 15: Charged Particles in Electric and Magnetic Fields

15

Notes

7. An electron beam in a cathode ray oscilloscope can be deflected by either an electric field or a magnetic field.
 - a) Compare these two methods with regard to
 - (i) the direction of the deflection produced;
 - (ii) the effect of the speed of the electrons on this deflecting force;
 - b) Show in a diagram how an electric field and a magnetic field can be arranged so that together they produce no resultant force on an electron passing through them.
 - c) Such a cancellation of forces can occur for only one particular electron velocity. Calculate the value of this velocity if the electric field has a strength of $1.00 \times 10^4 \text{ V m}^{-1}$ and the magnetic field strength is 0.100 tesla.

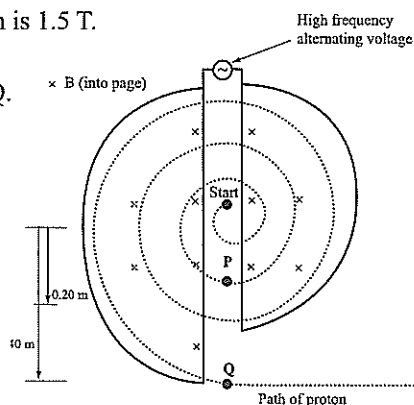
8. A proton is accelerated from rest through a potential difference of 20.0 kV. The proton then enters, at right angles, a uniform magnetic field of 0.200 T.
 - a) Determine the magnitude and direction of the acceleration of the proton in the magnetic field.
 - b) Explain why the resulting path of the proton is circular.
 - c) What is the strength of the electric field required to be superimposed on the magnetic field in order to allow the proton to proceed in a straight line?

9. An electron travelling at $1.60 \times 10^4 \text{ m s}^{-1}$ enters a region where there exists a uniform magnetic field of 3.00×10^{-2} tesla, at right angles to the electron's path.
 - a) Calculate the radius of curvature of the electron's path.

The electron now enters a region where the magnetic force exerted on it is balanced by an electric field.

 - b) Calculate the magnitude of the electric field strength required.
 - c) Show in a labelled diagram the vectors representing the motion of the electron, the magnetic field and the electric field.
 - d) A second electron having a velocity greater than $1.60 \times 10^4 \text{ m s}^{-1}$ enters the same region. Using labelled diagrams, show the paths of both electrons through the region, indicating the directions of both the magnetic and the electric fields.

10. A cyclotron accelerates small charged particles in a circular path to very high speeds, then releases the particles to strike a target and make radioisotopes. The cyclotron shown accelerates protons that start at its centre. These travel in circles because of two magnetic fields called dees. An alternating electric field in the gap between the dees accelerates the protons to higher velocities. The magnetic field strength is 1.5 T.
 - a) Why does the proton's path radius increase?
 - b) Calculate the proton velocities at positions P and Q.



- 4 a) (i) 2.4×10^{15} Hz; UV
(ii) 6.14×10^{14} Hz; visible
(iii) 4.56×10^{14} Hz; visible
(iv) 2.32×10^{14} Hz; IR
4 b) 4.58×10^{-19} J; E_5 to E_2
5 a) 45 keV
5 c) 30 keV; 40 keV
5 e) 2.0×10^{-16} J
7 a) 5.13 eV
7 b) 2.11 eV; E_2 to E_1
7 c) 0.002 eV
11 a) 60 keV (9.6×10^{-15} J)
11 b) 1.45×10^8 m s⁻¹
11 c) 2.07×10^{-11} m
12 12.4 kV
13 3.71×10^{-11} m
14 b) 4.14×10^{-11} m; 7.24×10^{18} Hz

Motion and Forces in Electric and Magnetic Fields

- Set 13 1 4.5×10^6 N C⁻¹
3 b) 2.94×10^7 m s⁻¹, deflected by 8.9°
3 e) 2.9×10^{-16} J
3 f) 6.0×10^4 V m⁻¹
5 a) 3.3×10^{-7} J
5 b) 66 V
6 100 V m⁻¹
7 a) 5.00 keV
7 b) 8.00×10^{-16} J
8 a) 10.0 keV
8 b) 1.6×10^{-15} J
9 a) 5.6×10^7 V m⁻¹
9 b) 2.4×10^{-15} J
10 b) 3.6×10^{-5} N
10 c) 10 mm; 0.090 V
13 3.91×10^5 m s⁻¹
14 b) 2500 eV (4.00×10^{-16} J)
14 c) 2.96×10^7 m s⁻¹
14 e) same
14 f) less
15 a) 4000 eV; 3.75×10^7 m s⁻¹
15 b) 2000 V m⁻¹; 3.20×10^{-16} N
16 b) 7.02×10^{15} m s⁻²
Set 14 7 1.58×10^{-26} kg to 1.62×10^{-26} kg
8 a) 41.8 m
8 b) 6.28×10^{-3} s
9 b) 2.4×10^{-14} N
10 b) 6.3×10^7 C kg⁻¹; 4.8×10^7 C kg⁻¹
Set 15 3 a) 1.10×10^5 V m⁻¹
3 b) Yes
4 c) They accelerate in a straight line at 1.76×10^{14} m s⁻²
5 a) 7.26×10^7 m s⁻¹
5 b) 1.76×10^{-4} m
5 c) 1.52×10^{-11} s